

Wavelet Kolmogorov–Arnold Networks for Inter-Patient ECG Heartbeat Classification: A Controlled Multi-Seed Evaluation

Venkateramanan Manivannan^a, Ramanathan Lakshmanan^{b,*}

^a*School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology, Vellore, 632014, Tamil Nadu, India*

^b*Department of IoT, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, 632014, Tamil Nadu, India*

Abstract

Kolmogorov–Arnold networks (KANs) with wavelet edge functions are promoted as inspectable models. They have not been evaluated for ECG heartbeat classification under the inter-patient protocol with repeated runs and matched baselines. We evaluated a 153,045-parameter wavelet KAN (PC-WavKAN) on MIT-BIH under the de Chazal DS1/DS2 partition against four baselines trained under the same protocol and budget. The design used 20 paired seeds, architecture selection on validation data only, Holm–Bonferroni correction, effect sizes and confidence intervals. DS2 Macro-F1 was 0.357 ± 0.019 and not statistically distinguishable from any baseline (all Holm-adjusted $p \geq 0.36$; every paired 95% interval within ± 0.03). Supraventricular recall was 0.198 at precision 0.175, and the supraventricular operating point was poor for all five models. On validation, a prior-centred initialisation of the wavelet parameters helped relative to an isotropic one, the mother wavelet did not measurably matter, and self-attention over RR-interval features was harmful. The trained parameters did not support a physiological interpretation: the channel groups are exchangeable, and training moved randomly initialised parameters away from the prior values. Class-balanced sampling raised held-out supraventricular recall from 0.137 to 0.222, but not on the validation partition. Re-evaluating all 100 checkpoints without retraining showed two

*Corresponding author.

Email addresses: `venkate.ramanan2024@vitstudent.ac.in` (Venkateramanan Manivannan), `lramanathan@vit.ac.in` (Ramanathan Lakshmanan)

things. An annotation-derived artefact in the RR features of 9.2% of test beats did not inflate performance. Under preprocessing matched to training, the wavelet KAN transferred worse than both convolutional baselines to the INCART and SVDB databases.

Keywords: Kolmogorov–Arnold networks, wavelets, ECG heartbeat classification, inter-patient evaluation, class imbalance, reproducibility

1. Introduction

The electrocardiogram (ECG) is the reference modality for arrhythmia detection, and manual review of long-term recordings does not scale. Convolutional and attention-based networks now report high beat-classification accuracy [1, 2, 3]. Much of that literature, however, evaluates under an intra-patient protocol in which beats from one recording appear in both training and test sets, which lets a classifier memorise a patient’s morphology and inflates reported accuracy [4, 5, 6]. Under the inter-patient DS1/DS2 protocol of de Chazal *et al.* [7], performance drops sharply and minority-class recall in particular collapses.

Kolmogorov–Arnold Networks (KANs) [8] place learnable univariate functions on network edges. Wav-KAN [9] uses wavelets parameterised by a translation and a dilation as those functions and argues that this makes the learned function space more inspectable. KAN-based ECG classifiers have been reported [10], as have learnable-wavelet convolutional networks for arrhythmia classification [11]. We found no wavelet-KAN evaluated under the de Chazal protocol with repeated seeds, protocol-matched baselines and multiplicity-corrected statistics. We also found no test of whether its parameters admit the physiological reading that the wavelet parameterisation invites.

This paper reports such an evaluation. It is framed as a controlled study, not as a method proposal. We ask four questions:

RQ1 Under the inter-patient protocol and a matched training protocol, does a wavelet KAN differ from standard architectures in five-class Macro-F1?

RQ2 Which design choices measurably matter, as judged on validation data alone: the initialisation of the wavelet parameters, the choice of mother wavelet, the rhythm encoder, an auxiliary morphology branch, and class-balanced sampling?

29 **RQ3** Do the trained wavelet parameters support a physiological interpreta-
30 tion?

31 **RQ4** How sensitive are the conclusions to protocol artefacts and to a change
32 of database?

33 The evaluated model, PC-WavKAN (153,045 parameters), is a wavelet-
34 KAN morphology branch combined with an interval-feature (rhythm) branch.
35 Its one non-standard ingredient is a prior-centred initialisation of the wavelet
36 translations and dilations (PCWI). Our contributions are empirical:

- 37 1. A 20-seed, seed-paired inter-patient comparison against four baselines
38 trained under the same protocol and budget, with architecture selec-
39 tion on validation data only, Holm–Bonferroni correction, effect sizes
40 and confidence intervals. It finds no statistically detectable Macro-F1
41 difference. Every paired 95% confidence interval includes zero and lies
42 within ± 0.03 , and the S-class operating point is poor for all five models
43 (§4.2, §4.3).
- 44 2. A single-variable component analysis on validation data (§4.1). Initialis-
45 ing the wavelet parameters around fixed prior values (PCWI) improves
46 validation Macro-F1 relative to an isotropic initialisation in the config-
47 uration in which the components were ablated. The mother wavelet
48 is not measurably consequential. Self-attention over the five interval
49 features is harmful and was removed.
- 50 3. A test of the interpretability premise with explicit null models (§4.7).
51 The channel blocks that receive the three priors are exchangeable, so
52 no block can be assigned a waveform component. Training preserves
53 much of the prior-centred structure, but it moves randomly initialised
54 parameters *away* from the prior values, so the parameters do not support
55 a physiological reading.
- 56 4. Inference-only sensitivity analyses of all 100 checkpoints.
 - 57 • An annotation-derived artefact in the interval features does not
58 inflate performance (§4.6).
 - 59 • Under a preprocessing pipeline verified identical to training, the
60 wavelet KAN transfers worse than both convolutional baselines to
61 two external databases (§4.8). An unmatched evaluation of the
62 same checkpoints had suggested the opposite.

63 A further finding concerns class-balanced sampling. Compared with the
64 inverse-frequency reweighting used by the baselines, it raised DS2 supraven-

65 tricular recall substantially in two configurations, but this gain did not
66 reproduce on the validation partition for the adopted configuration (§4.4).

67 2. Related Work

68 *Inter-patient evaluation.* Systematic reviews document how prevalent intra-
69 patient evaluation remains and how strongly it inflates reported performance
70 [5, 6]. Bahrami *et al.* quantify that optimism directly [4]. The DS1/DS2
71 partition of de Chazal *et al.* [7] is the established remedy for MIT-BIH, and
72 we adopt it unmodified. Architectures evaluated under it include CNN–RNN
73 hybrids [2], multiscale attention CNNs [12] and sequence-to-sequence models
74 [13]. Differences in lead selection, beat extraction, AAMI mapping and Q-class
75 handling make published accuracies hard to compare directly, so we retrain
76 our baselines under one protocol (§3.5). We give the most direct external
77 comparison, with the protocol’s own source, in §4.3.

78 *KANs and learnable wavelet parameterisations.* KANs [8] use B-spline edge
79 functions. Wav-KAN [9] substitutes wavelets, including the Mexican Hat,
80 with learnable translation and dilation. The wavelet-KAN layer, the Mexican
81 Hat basis and the argument that a wavelet basis is more inspectable are due
82 to that work and are not claimed here. Learning only the scale and translation
83 of wavelet kernels has also been used in convolutional front-ends, for example
84 WaveletKernelNet [14] for machinery fault diagnosis and WavelNet [11], a
85 SincNet-inspired wavelet CNN for arrhythmia classification on a subject-
86 oriented MIT-BIH benchmark. Unlike those convolutional kernels, a Wav-
87 KAN edge function acts on the *amplitude* of an individual input sample,
88 not on a time window, a distinction that matters for interpretation (§3.3).
89 MAK-Net [10] combines spline KAN layers with multiscale convolution and a
90 BiGRU for imbalanced ECG classification. It evaluates under an intra-patient
91 protocol and reports Grad-CAM attributions. Initialisation of KANs is an
92 active question: Rigas *et al.* [15] compare variance-preserving and power-law
93 schemes for spline-KAN coefficients. PCWI addresses a different quantity,
94 the wavelet translation and dilation, and leaves coefficient initialisation at a
95 standard default.

96 *Class imbalance.* In a systematic study on image benchmarks, Buda *et al.*
97 [16] found oversampling to be the dominant remedy for class imbalance in
98 convolutional networks. They compared it with undersampling, two-phase

99 training and threshold adjustment. Kang *et al.* [17] compare instance-
 100 balanced, class-balanced and progressively balanced sampling for long-tailed
 101 recognition. Curriculum approaches vary the sampling distribution or loss
 102 weighting over training [18, 19, 20], and adaptive logit adjustment is a logit-
 103 level alternative studied for record-level ECG diagnosis [21]. Our training
 104 script implements a two-phase schedule of this kind. The selected checkpoint
 105 always came from its first, class-balanced phase (§3.4), so what our comparison
 106 measures is class-balanced sampling against inverse-frequency reweighting.

107 *Efficient inference.* Small memory footprints are argued for on-device ECG
 108 inference [22, 23]. We report measured size and x86 CPU latency, and say
 109 which hardware class was measured (§4.9).

110 3. Data and Methods

111 3.1. Data and Preprocessing

112 We use the 44 non-paced records of the MIT-BIH Arrhythmia Database
 113 [24, 25]. The paced records 102, 104, 107 and 217 are excluded under the
 114 protocol. Channel 0 is used, which is the modified lead II (MLII) in every
 115 record except 114, where channel 0 is V5. Each record is cleaned over its full
 116 length with the `neurokit` method of NeuroKit2, a zero-phase 0.5 Hz fifth-
 117 order Butterworth high-pass filter followed by a zero-phase 7-sample moving
 118 average, which the library designs for 50 Hz mains. A 360-sample window
 119 (1 s at 360 Hz) is then extracted at each annotated beat. The window spans
 120 90 samples (250 ms) before and 270 samples (750 ms) after the annotated
 121 R-peak, so the R-peak sits at sample 90. Each window is z -scored with its
 122 own mean and standard deviation, and windows with degenerate variance
 123 are discarded. No normalisation statistic is estimated across beats, records or
 124 splits.

125 *Interval features.* Each beat i also receives a five-element vector

$$r_i = [RR_{-2}, RR_{-1}, RR_0, RR_{+1}, RR_{+2}], \quad (1)$$

126 where RR_0 is the interval ending at beat i , RR_{-1} and RR_{-2} are the two
 127 intervals before it, and RR_{+1} and RR_{+2} are the two intervals after it. Intervals
 128 are in seconds, clipped to $[0.2, 3.0]$ s, and set to 0.8s where they would
 129 extend past the start or end of a record. They are computed between
 130 consecutive entries of the database annotation file. Those entries include

131 non-beat annotations such as rhythm-change, noise and artefact markers, so
 132 a small fraction of intervals are annotation-to-beat gaps rather than beat-to-
 133 beat intervals. §4.6 quantifies this artefact and its effect on every reported
 134 model. It affects all models identically, because every model receives the same
 135 features.

136 *Temporal scope.* Three properties make the task *offline* beat classification
 137 rather than streaming classification. The interval vector contains two intervals
 138 that follow beat i . The window’s 750 ms post-R segment contains the next
 139 annotated beat for 48.8% of DS2 beats. The filter is zero-phase and applied
 140 per record. We therefore do not describe the model as real-time or streaming.
 141 A deployment would need a lookahead buffer and a causal filter.

142 3.2. Inter-Patient Split

143 We use the DS1/DS2 partition of de Chazal *et al.* [7], splitting DS1
 144 into training and validation records following Takalo-Mattila *et al.* [1]. DS1
 145 training uses the 18 records 101, 106, 108, 109, 112, 114, 115, 116, 118, 119,
 146 122, 124, 201, 203, 205, 207, 215 and 220. DS1 validation uses the 4 records
 147 208, 209, 223 and 230, which carry every model and architecture selection
 148 decision. DS2, held out, comprises the 22 records 100, 103, 105, 111, 113,
 149 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232,
 150 233 and 234. The partition is by record, and record disjointness is asserted
 151 programmatically and covered by a regression test. One subject contributes
 152 to two partitions: records 201 (training) and 202 (DS2) were taken from the
 153 same analogue tape, a known property of the de Chazal partition that we
 154 retain unmodified.

155 Annotations are mapped to the five AAMI EC57 classes [5, 26] (Table 1).
 156 The Q class has only 15 beats in these records (6 in training, 7 in DS2), so
 157 no per-class conclusion can be drawn for it. Record 208 holds 372 of DS1’s
 158 414 fusion beats and lies in the validation partition, which leaves 28 F beats
 159 for training.

160 3.3. PC-WavKAN

161 The model (Fig. 1) has a morphology branch and a rhythm branch whose
 162 outputs are concatenated and classified.

Table 1: Class distribution and AAMI mapping.

Class	Included Annotations	Train	Val	Test (DS2)	Total	Ratio (vs N)
N	Normal, L, R, e, j	37,337	8,502	44,232	90,071	1.0 : 1
S	A, a, J, S	485	458	1,837	2,780	32.4 : 1
V	V, E	2,321	1,467	3,220	7,008	12.8 : 1
F	F	28	386	388	802	112.3 : 1
Q	/, f, Q	6	2	7	15	6004 : 1
Total		40,177	10,815	49,684	100,676	

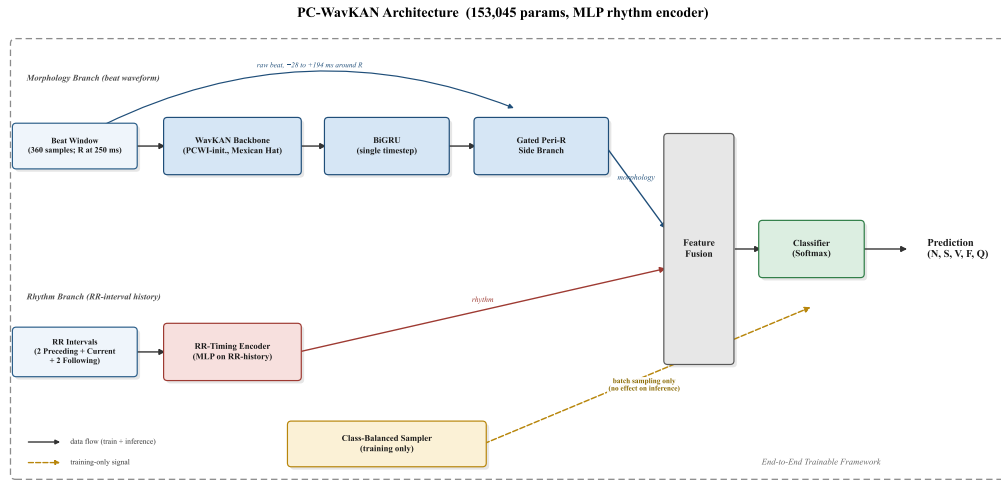


Figure 1: PC-WavKAN as implemented. The morphology branch passes the beat window through a wavelet-KAN layer with prior-centred initialisation (PCWI), a bidirectional GRU applied to a single time step, and a gated side branch that reads raw-beat samples 80–160 (28 ms before to 194 ms after the R-peak). The rhythm branch encodes the five interval features with an MLP. The class-balanced sampler affects training only.

163 *Wavelet-KAN layer.* Following Wav-KAN [9], every edge (j, k) carries its
 164 own Mexican Hat function $\psi(u) = (1 - u^2)e^{-u^2/2}$ (Fig. 2). The layer maps
 165 $x_i \in \mathbb{R}^{360}$ to $\mathbb{R}^{64}$:

$$z^{(k)} = \sum_{j=1}^{360} w_{j,k} \psi\left(\frac{x_j - \mu_{j,k}}{|\gamma_{j,k}|}\right) + \lambda \sum_{j=1}^{360} v_{j,k} x_j, \quad k = 1, \dots, 64, \quad (2)$$

166 with learnable translation $\mu_{j,k}$, dilation $\gamma_{j,k}$ and weight $w_{j,k}$, and a linear
 167 residual path with fixed gate $\lambda = 0.1$. ψ is evaluated on the *amplitude* x_j of
 168 input sample j , so μ and γ are an amplitude offset and an amplitude width.
 169 They are not a temporal position or duration. The output is layer-normalised,

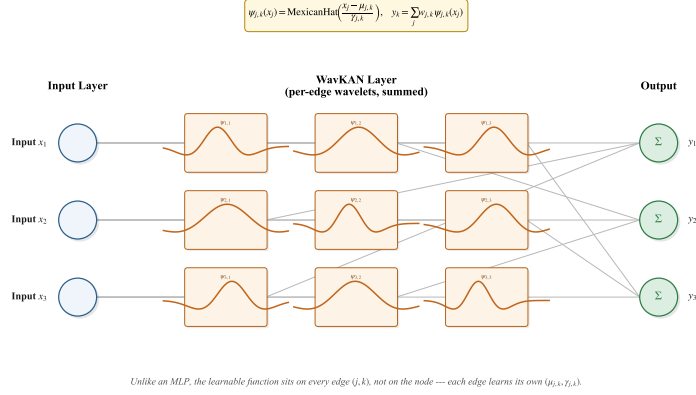


Figure 2: Wavelet-KAN edge parameterisation. Every edge of the $j \times k$ grid carries its own translation and dilation, as in (2).

170 passed through dropout ($p = 0.2$), and fed to a bidirectional GRU with 32
 171 units per direction. The 64 activations form a single token, so the GRU
 172 executes one step in each direction from a zero state. It therefore acts as
 173 a gated transformation of the channel vector and integrates no temporal
 174 context. It has 18,816 parameters (12.3% of the model).

175 *Prior-centred wavelet initialisation (PCWI)*. The 64 output channels are split
 176 into three blocks, and every edge feeding block c is initialised around a fixed
 177 pair $(\bar{\mu}_c, \bar{\gamma}_c)$:

$$\mu_{j,k} \sim \mathcal{U}(\bar{\mu}_c - \epsilon_c, \bar{\mu}_c + \epsilon_c), \quad \gamma_{j,k} \sim \mathcal{U}(\bar{\gamma}_c - \delta_c, \bar{\gamma}_c + \delta_c). \quad (3)$$

178 Channels 1–32 use $(0.0, 0.05)$ with $(\epsilon, \delta) = (0.02, 0.005)$, channels 33–48
 179 use $(-0.10, 0.12)$, and channels 49–64 use $(+0.10, 0.10)$, the last two with
 180 $(\epsilon, \delta) = (0.03, 0.01)$. The values were chosen with reference to the relative
 181 amplitude and sharpness of the QRS complex and the P and T waves on
 182 the z -scored scale, and we refer to the blocks by those names (QRS-, P- and
 183 T-prior blocks). Without PCWI, all edges use the isotropic initialisation
 184 $\mu \sim \mathcal{U}(-0.3, 0.3)$, $\gamma \sim \mathcal{U}(0.05, 0.20)$. The weights $w_{j,k}$ and $v_{j,k}$ use Kaiming-
 185 uniform initialisation in both cases.

186 Two properties limit what PCWI can encode. First, each block’s prior
 187 is shared by all 360 input samples, so no prior is tied to the temporal
 188 location of a waveform component. Second, the 64 channels are exchangeable.

189 They feed a layer normalisation initialised identically across channels and a
190 recurrent layer whose input weights are drawn independently and identically
191 across channels. Permuting the channels together with the corresponding
192 downstream weights leaves the network function unchanged, which we verified
193 numerically (maximum output difference 10^{-7}). A network in which the P-
194 and T-block priors are exchanged is therefore an exact relabelling of one in
195 which they are not, with the same initial distribution. What PCWI determines
196 is the set of initial (μ, γ) values, 32 channels narrow and centred at zero and
197 16 each slightly offset and wider, not an assignment of channels to waveform
198 components. We name it *prior-centred* for that reason.

199 *Gated side branch.* A side branch reads raw-beat samples 80–160, which with
200 the R-peak at sample 90 spans 28 ms before to 194 ms after the R-peak, i.e.
201 the QRS complex and early ST segment. The branch was designed to cover
202 the P-wave, but its sample range was fixed for an R-peak at sample 180,
203 and it does not reach the P-wave in the window as extracted. We therefore
204 describe it by what it reads. The samples pass through a randomly initialised
205 wavelet-KAN layer ($80 \rightarrow 32$), are projected to 64 dimensions, and pass
206 through a four-head attention block over a single key/value token. With
207 one token the attention softmax is degenerate, so at inference the block is a
208 learned linear map. A sigmoid gate then combines the result with the GRU
209 output. The branch has 34,816 parameters.

210 *Rhythm branch and classifier.* The interval vector is encoded to 16 dimensions
211 by an MLP ($5 \rightarrow 64 \rightarrow 32 \rightarrow 16$, ReLU). An alternative encoder that projects
212 each interval to a 16-dimensional token and applies two-head self-attention
213 was also evaluated (§4.1). The 80-dimensional concatenation is classified by
214 an $80 \rightarrow 48 \rightarrow 5$ head. With the MLP rhythm encoder the model has 153,045
215 trainable parameters, and 154,325 with the self-attention encoder. All results
216 outside §4.1 use the 153,045-parameter configuration.

217 3.4. Training

218 Training uses AdamW (learning rate 10^{-3} , weight decay 10^{-4}), batch
219 size 64, gradient-norm clipping at 1.0 and a cosine learning-rate schedule
220 over a nominal 100-epoch budget. Early stopping monitors DS1-validation
221 Macro-F1 with patience 15, and the checkpoint of the best validation epoch
222 is retained. In the PC-WavKAN runs trained with class-balanced sampling,
223 training stopped between epochs 17 and 40, so the learning rate changed little

224 before the retained epoch. The baseline scripts did not log their stopping
225 epoch.

226 *Class-balanced sampling.* For the first 25 epochs, mini-batches are drawn
227 by a sampler that selects each class with equal probability, and the loss is
228 unweighted cross-entropy. The training script then enters a second phase in
229 which the sampler is annealed towards the natural class distribution and the
230 loss becomes a class-weighted focal loss. In all 60 runs that used this schedule
231 (three configurations, 20 seeds each), the retained checkpoint came from the
232 first phase (best epochs 2–25), because validation Macro-F1 never improved
233 after the switch. The second phase therefore did not contribute to any
234 reported model. We describe and evaluate the procedure by what determined
235 those models: class-balanced sampling with unweighted cross-entropy and
236 validation-based checkpoint selection, abbreviated CBS. The comparator,
237 used by all four baselines and by the no-CBS arms, is natural-distribution
238 sampling with inverse-frequency class-weighted cross-entropy, with the S-class
239 weight multiplied by 8.

240 *Augmentation.* During training only, every non-N beat in a mini-batch is
241 perturbed with probability one. The perturbation is additive Gaussian noise
242 at 25 dB SNR, a sinusoidal baseline wander of 0.5–2.5 Hz with amplitude
243 0.01–0.05 (in z -units), and amplitude scaling by $\mathcal{U}(0.9, 1.1)$, applied after
244 z -scoring. The baselines are trained without augmentation (§3.5).

245 3.5. Baselines and What Is Matched

246 Four architectures spanning 29K–118K parameters are compared (Table 2).
247 *Held identical:* data, preprocessing, split, input representation (every model
248 receives both the beat window and the interval vector), optimiser, learning
249 rate, weight decay, schedule, epoch budget, early-stopping rule and patience,
250 batch size, gradient clipping, checkpoint-selection rule, the 20 seeds, and
251 the class-weight formula. *Not held identical:* the imbalance mechanism and
252 augmentation. The baselines use natural-distribution sampling with class-
253 weighted cross-entropy, except CNN+Focal, which uses focal loss without
254 static class weights. We also tested a class-balanced sampler combined with
255 inverse-frequency weights for the baselines. At this imbalance ratio that
256 combination drove the majority-class gradient to near zero, and three of
257 the four baselines then never predicted class N, so it was not used. The
258 comparison in §4.2 is therefore between complete methods, i.e. architecture

Table 2: Baselines. Protocol, split, inputs, optimiser, budget, checkpoint rule and seeds are identical to PC-WavKAN (§3.5).

Model	Params	Imbalance handling
ResNet1D (4-block residual CNN, 32 channels)	62,869	Weighted CE
Transformer ($d_{\text{model}}=32$, 2 layers, 4 heads)	29,653	Weighted CE
CNN+Focal (4-layer 1D CNN)	71,061	Focal loss ($\gamma=2$)
B-Spline KAN (single B-spline per edge; no PCWI or side branch)	118,229	Weighted CE
PC-WavKAN (§3.3)	153,045	CBS + augmentation

259 together with its imbalance strategy, under a common protocol and budget.
 260 §4.4 and §4.5 quantify the two mechanisms on PC-WavKAN itself.

261 The baseline labelled B-Spline KAN is the same wavelet-KAN layer with
 262 a single cubic B-spline in place of the Mexican Hat on every edge, isotropic
 263 initialisation, no side branch, and the same GRU, rhythm MLP and head. It
 264 is not the grid-based spline KAN of Liu *et al.* [8].

265 3.6. Parameter-Space Prior Retention

266 To test RQ3 we measure how close the trained wavelet parameters remain
 267 to the PCWI prior values. For block c ,

$$\text{PPR}_c = \max\left(0, 1 - \frac{1}{2} \left[\frac{|\overline{\mu_{j,k}} - \bar{\mu}_c|}{\Delta_\mu} + \frac{||\overline{\gamma_{j,k}}| - \bar{\gamma}_c|}{\Delta_\gamma} \right] \right), \quad (4)$$

268 where the overlines denote means over all edges of block c , and $\Delta_\mu = 0.40$,
 269 $\Delta_\gamma = 0.20$ are fixed normalising spans. The reported PPR is the unweighted
 270 mean over the three blocks. The measure is computed from parameters alone
 271 and never observes a model output. Because PCWI initialises at the prior
 272 values, an untrained PCWI model scores near-maximally by construction.
 273 PPR is therefore interpretable only against reference points, and we report
 274 three: untrained with PCWI, untrained with isotropic initialisation, and
 275 *trained* with isotropic initialisation. The last one tests whether training
 276 moves parameters towards the prior values. Per-edge absolute deviations
 277 are averaged rather than block means, because per-edge drift is close to
 278 symmetric and largely cancels in the mean (in the trained model, mean
 279 per-edge $|\Delta_\mu| = 0.064$ against 0.004 for the deviation of the block means).

280 3.7. Statistical Methodology

281 Every arm is trained with the same 20 seeds: 42, 101, 777, 2026, 9999,
 282 1234, 2024, 31415, 27182, 7, 11, 13, 888, 555, 333, 99, 1001, 5050, 8080

and 1998. Comparisons are paired by seed label. Within one architecture, the same seed gives the same initialisation. Across architectures the pairing is by label only and implies no shared initialisation or data order. The signed-rank test remains valid in that case under the null hypothesis of identical distributions. Significance is assessed by the two-sided Wilcoxon signed-rank test [27], with Holm–Bonferroni correction [28] within declared families. The metrics of one arm-versus-arm comparison form one family, as do the comparisons against the four baselines on one metric. All p -values are Holm-adjusted unless labelled raw. Effect sizes are the paired standardised mean difference $d_z = \bar{\delta}/s_\delta$, reported against Cohen’s conventional thresholds [29]. **Positive d_z favours PC-WavKAN, the adopted configuration, or the treatment arm.** We also report 95% confidence intervals on the paired mean difference. At $n = 20$ and $\alpha = 0.05$ two-sided, a paired test has 80% power only for $d_z \gtrsim 0.66$, and non-significant results are reported with their intervals rather than as evidence of absence. No equivalence margin was specified, so parity results are not equivalence claims.

Outcome hierarchy. Results are reported in §4. The *primary* outcome is DS2 five-class Macro-F1 of the adopted configuration against the four baselines. The *pre-specified secondary* outcome is the effect of the imbalance mechanism on per-class recall, tested on the base configuration (§4.4). Everything else is *exploratory*. That includes the same comparison on the adopted configuration, which uses arms trained for the augmentation analysis, and the interval ablation, the cross-database evaluation, the prior-retention analysis and the artefact sensitivity analysis. The last four, and the four-class Macro-F1, were added during a final audit after the primary results were known, and are labelled as such where reported.

Software. Models are implemented in PyTorch. AI-assisted coding (Claude, Anthropic) was used to write and audit parts of the preprocessing, evaluation, statistics and figure code. Every quantitative claim in this paper is re-derived from the per-seed result files by a verification script released with the code (Code availability).

4. Results

4.1. Architecture Selection on Validation Data

The configuration reported in the rest of this paper was selected using DS1 validation only. Table 3 gives the single-variable ablation. The base configuration has the wavelet-KAN layer with PCWI, the gated side branch,

Table 3: Single-variable component ablation on **DS1 validation**, 20 seeds per row. Base configuration: PCWI, gated side branch, self-attention rhythm encoder and CBS; validation Macro-F1 0.4145 ± 0.0201 . Δ and d_z are variant minus base, so positive values favour the variant. p is Holm-adjusted across the seven comparisons.

Variant	Val. Macro-F1	Δ vs base	d_z	Holm p	Reading
No PCWI (isotropic init.)	0.3932 ± 0.0129	-0.0213	-1.05 (large)	0.0001	PCWI helps in this configuration
No gated side branch	0.3976 ± 0.0184	-0.0169	-0.54 (medium)	0.164	not resolved
No CBS (reweighting)	0.4053 ± 0.0173	-0.0092	-0.37 (small)	0.624	not resolved
Basis: Morlet	0.4219 ± 0.0187	$+0.0074$	$+0.31$ (small)	0.568	not resolved
Basis: DOG	0.4041 ± 0.0123	-0.0104	-0.45 (small)	0.164	not resolved
Basis: B-spline	0.4116 ± 0.0151	-0.0029	-0.14 (negl.)	0.624	not resolved
MLP rhythm encoder	0.4397 ± 0.0140	$+0.0252$	$+1.15$ (large)	0.0008	adopted

the self-attention rhythm encoder and CBS, and each row changes exactly one component. The validation score of each seed is taken at the epoch its checkpoint was saved.

Three findings follow. *Initialisation*: replacing PCWI with the isotropic initialisation costs 0.021 validation Macro-F1 ($d_z = -1.05$, Holm $p = 1 \times 10^{-4}$), the only corrected positive effect among the components ablated. The comparison contrasts one structured initialisation with one isotropic distribution, and §3.3 shows that what PCWI fixes is a set of amplitude offsets and widths, not a physiological assignment. The finding is therefore that this initialisation helps, not that physiological content does. It was measured in the base configuration and was not re-measured on the adopted configuration. *Mother wavelet*: substituting Morlet, DOG or a B-spline for the Mexican Hat changes validation Macro-F1 by at most 0.010, and no difference approaches significance. *Rhythm encoder*: replacing self-attention with an MLP over the same five intervals improves validation Macro-F1 by 0.025 ($d_z = +1.15$, Holm $p = 8 \times 10^{-4}$). This configuration is the best of all eight on validation. It outperforms each of the seven alternatives (Holm-adjusted $p \leq 0.0032$ over that family), and we adopt it.

Test-set exposure. Table 3 alone determines the selection. DS2 metrics for non-adopted variants were nevertheless computed during development, and the DS2 ranking agrees with the validation ranking (adopted 0.357 ± 0.019 against 0.323 ± 0.018 for the base). DS2 was therefore not untouched over the course of the study, and §4.2 should be read as a held-out estimate under a validation-determined selection rule rather than a single-shot test. Its conclusion is a null result, which selection on DS2 would not be expected to produce. On arrays regenerated from PhysioNet with the released code, all 20 adopted-configuration checkpoints reproduce their DS2 predictions beat for

Table 4: **Primary outcome.** DS2 five-class Macro-F1, PC-WavKAN versus four baselines, 20 paired seeds. Δ , CI and d_z are PC-WavKAN minus baseline; p is Holm-adjusted across the four comparisons.

Model	Params	Macro-F1	Δ	95% CI of Δ	d_z / Holm p
ResNet1D	62,869	0.351 ± 0.019	+0.006	$[-0.008, +0.020]$	+0.19 / 1.00
Transformer	29,653	0.353 ± 0.040	+0.004	$[-0.013, +0.022]$	+0.12 / 1.00
CNN+Focal	71,061	0.362 ± 0.015	-0.005	$[-0.017, +0.006]$	-0.22 / 1.00
B-Spline KAN	118,229	0.371 ± 0.024	-0.014	$[-0.029, +0.001]$	-0.43 / 0.36
PC-WavKAN	153,045	0.357 ± 0.019	—	—	—

beat, and 19 reproduce exactly the validation Macro-F1 logged at their saved epoch. Seed 1001’s logged validation score does not reproduce. Excluding it changes no corrected conclusion, though the unadjusted interval against B-Spline KAN (Table 4) then narrowly excludes zero ($[-0.031, -0.0002]$).

4.2. Primary Comparison

PC-WavKAN’s DS2 Macro-F1 is not statistically distinguishable from any baseline (Table 4). All Holm-adjusted $p \geq 0.36$, all effects are negligible to small, and every 95% interval on the paired difference contains zero, the widest bound being 0.029. The data exclude a large difference in either direction. They do not establish equivalence, for which no margin was set. PC-WavKAN is also the largest of the five models. Because the Q class contributes a near-constant zero (§4.3), we also report the four-class N/S/V/F Macro-F1 (exploratory): PC-WavKAN 0.446 ± 0.023 , ResNet1D 0.439 ± 0.023 , Transformer 0.441 ± 0.050 , CNN+Focal 0.453 ± 0.019 and B-Spline KAN 0.463 ± 0.030 , with all Holm-adjusted $p \geq 0.36$. The conclusion is unchanged.

Fig. 3 shows the structure behind these values for one seed. N and V separate well, each leaking under 6% into the other, with V-recall 0.905. S beats are assigned predominantly to V (0.42) and secondarily to N (0.30), and F beats are almost entirely predicted as N (0.87).

4.3. Per-Class Operating Points

Table 5 gives precision, recall and F1 per class over 20 seeds, recomputed from stored per-seed predictions. The per-class F1 values average to the Macro-F1 of Table 4 for all five models.

The S-class operating point is poor in both directions. Recall is 0.198 at precision 0.175, so roughly four in five predicted S beats are wrong and

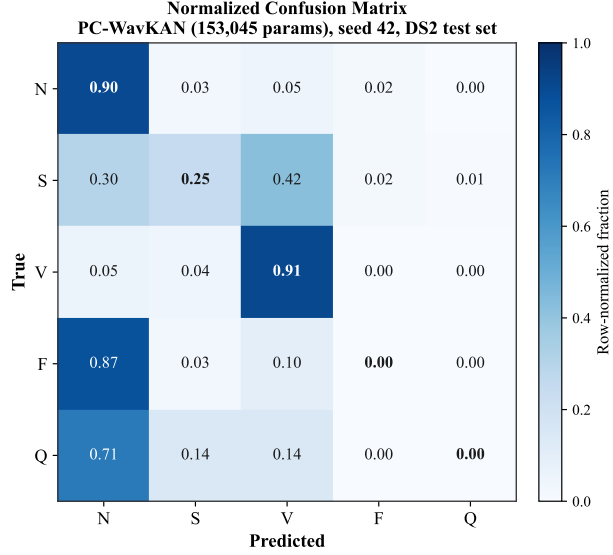


Figure 3: Row-normalised DS2 confusion matrix, PC-WavKAN, seed 42, generated from this checkpoint’s own predictions.

Table 5: Per-class precision, recall and F1, 20 seeds, DS2. **(a)** PC-WavKAN, all classes. **(b)** S class across models, with the Holm-adjusted paired comparison of S-F1 against PC-WavKAN (positive d_z favours PC-WavKAN). S-class differences between models combine architecture and imbalance mechanism (§3.5). Q scores exactly zero for every model.

(a) PC-WavKAN, per class				
Class	Support	Precision	Recall	F1
N	44,232	0.971 ± 0.004	0.905 ± 0.025	0.936 ± 0.013
S	1,837	0.175 ± 0.079	0.198 ± 0.067	0.180 ± 0.064
V	3,220	0.528 ± 0.051	0.898 ± 0.020	0.663 ± 0.039
F	388	0.005 ± 0.006	0.006 ± 0.006	0.005 ± 0.006
Q	7	0.000	0.000	0.000
(b) S class across models				
Model	S-Prec.	S-Rec.	S-F1	d_z / Holm p
PC-WavKAN	0.175	0.198	0.180	—
ResNet1D	0.173	0.100	0.124	+0.77 / 0.004
Transformer	0.219	0.229	0.207	−0.35 / 0.312
CNN+Focal	0.306	0.075	0.115	+0.85 / 0.002
B-Spline KAN	0.237	0.280	0.244	−0.66 / 0.027

371 four in five true S beats are missed. On S-F1, PC-WavKAN is better than
 372 ResNet1D and CNN+Focal, indistinguishable from the Transformer, and
 373 worse than B-Spline KAN ($d_z = -0.66$, Holm $p = 0.027$). The parity of
 374 §4.2 is therefore not uniform across classes. Q contributes an exact zero for
 375 every model, since 6 training beats cannot support a learned class. This caps
 376 the attainable five-class Macro-F1 at 0.8 and should be recalled whenever
 377 five-class inter-patient Macro-F1 values are compared.

378 *Comparison with the protocol’s source.* de Chazal *et al.* [7], whose partition
 379 we adopt, report SVEB sensitivity 0.759 at positive predictivity 0.385, and
 380 VEB sensitivity 0.777 at positive predictivity 0.819. On the S class their
 381 linear-discriminant system is substantially better than every model trained
 382 here on both axes. On the V class our recall of 0.898 exceeds theirs, while
 383 our precision of 0.528 is lower. Their system uses two leads, hand-engineered
 384 morphology at fiducial points and interval features normalised to a record-level
 385 average, where ours uses one lead and a fixed window. We have not rerun
 386 their method. We report the comparison because the S-class gap is large and
 387 is against the protocol’s own source.

388 4.4. Class-Balanced Sampling Versus Reweighting

389 The pre-specified test compares CBS with reweighting on the base config-
 390 uration, with augmentation in both arms. S-recall rises from 0.079 ± 0.016
 391 to 0.123 ± 0.052 (Holm $p = 0.0016$ across five metrics, $d_z = +0.77$). On
 392 DS1 validation it rises from 0.218 ± 0.048 to 0.305 ± 0.047 ($d_z = +1.36$, raw
 393 $p = 2.4 \times 10^{-4}$). Table 6 repeats the comparison on the adopted configuration.
 394 It uses the two augmentation-free arms trained for §4.5, so the analysis is
 395 exploratory.

396 On DS2, CBS raises S-recall from 0.137 to 0.222 (95% CI of the difference
 397 $[+0.058, +0.112]$). The absolute F-recall gain is negligible ($0.0003 \rightarrow 0.0022$).
 398 V- and N-recall show no corrected change. On the validation partition the
 399 direction reverses. CBS is nominally worse on both validation Macro-F1 and
 400 S-recall, although after Holm correction across the two validation metrics
 401 neither difference is significant ($p \approx 0.05$). The DS2 gain in S-recall is therefore
 402 robust across two configurations but is not reproduced on the four-record
 403 validation partition for the adopted configuration. We read this as evidence
 404 that the benefit of class-balanced sampling for supraventricular beats depends
 405 on the patients evaluated, and we do not claim it as a general property. The

Table 6: CBS versus reweighting on the adopted configuration, both arms without augmentation, 20 paired seeds (exploratory). DS2 rows: d_z and p are CBS minus reweighting, Holm-adjusted across the five metrics. Validation rows: values at each run’s retained epoch, raw p .

Metric	Reweighting	CBS	d_z	p
<i>DS2 (held out)</i>				
Macro-F1	0.351 ± 0.014	0.364 ± 0.021	+0.52 (medium)	0.098
S-Recall	0.137 ± 0.037	0.222 ± 0.057	+1.47 (large)	2.9×10^{-5}
F-Recall	0.0003	0.0022	+0.88 (large)	0.013
V-Recall	0.880 ± 0.024	0.880 ± 0.017	−0.00 (negl.)	0.756
N-Recall	0.927 ± 0.021	0.917 ± 0.024	−0.32 (small)	0.228
<i>DS1 validation</i>				
Macro-F1	0.4285 ± 0.0125	0.4195 ± 0.0098	−0.57 (medium)	0.027 (raw)
S-Recall	0.2639 ± 0.0510	0.2391 ± 0.0425	−0.53 (medium)	0.025 (raw)

pre-specified test on the base configuration is the confirmatory evidence, and it is modest in absolute terms: an S-recall of 0.123.

4.5. Augmentation

PC-WavKAN is trained with minority-class augmentation and the base-lines are not. On validation, augmentation is supported: with CBS in both arms, it raises validation Macro-F1 by 0.020 ($d_z = +1.16$, raw $p = 1.7 \times 10^{-4}$), consistent with its place in the adopted recipe. On DS2, removing it leaves Macro-F1 statistically unchanged (0.357 with augmentation and 0.364 without, Holm $p = 0.46$) and reduces F-recall (0.0061 to 0.0022, Holm $p = 0.043$). The augmentation asymmetry therefore does not explain the parity in Table 4. Under an isolated 60-epoch recipe without CBS, the three noise-based strategies raised V-recall and lowered Macro-F1, and SMOTE produced the largest S-recall gain (0.316 $\rightarrow$ 0.648) at the largest Macro-F1 cost. That penalty did not carry over to the full recipe, which cautions against extrapolating short single-mechanism ablations.

4.6. Interval Features, Lookahead and an Annotation-Derived Artefact

Table 7 reports an exploratory leave-one-out analysis in which each interval position is masked at inference. It was run on the base configuration, whose rhythm features are identical to the adopted one’s.

Masking RR_0 causes by far the largest degradation, as expected, since a shortened preceding interval is the classical sign of a premature beat. The

Table 7: Leave-one-out interval ablation, base configuration, 20 seeds, DS2, exploratory. Baseline S-recall with the full vector: 0.1225 ± 0.0524 . p is Holm-adjusted across the five positions.

Masked position	Δ S-Recall	Holm p
RR_{-2}	-0.0020 ± 0.0077	0.221
RR_{-1}	-0.0016 ± 0.0075	0.221
RR_0	-0.0323 ± 0.0492	3×10^{-4}
RR_{+1}	$+0.0022 \pm 0.0095$	0.358
RR_{+2}	-0.0045 ± 0.0103	0.014

effect of masking RR_{+2} is small but survives correction. The model therefore draws signal from an interval that has not elapsed when beat i occurs, which is one reason we treat the task as offline (§3.1).

Annotation-derived artefact. Because the intervals are computed between consecutive annotation-file entries, including non-beat markers, 4,578 of 49,684 DS2 beats (9.2%) have at least one interval that differs from the beat-to-beat value, and 1,433 (2.9%) have a different RR_0 . By class, RR_0 differs for 6.4% of V beats, against 2.7% of N beats and 1.3% of S beats, and the recorded value is almost always shorter than the true interval, which mimics prematurity. A rhythm-change marker is placed where an arrhythmic episode begins, so this artefact carries annotation-derived information that would not be available on unannotated recordings, and it correlates with the label. We quantified its effect without retraining. We recomputed every DS2 interval between beat annotations only and re-evaluated all 100 published checkpoints (five models, 20 seeds). The recomputation leaves the beats and their order unchanged. As a check, the same re-evaluation with the published intervals reproduced every model’s reported Macro-F1. Of the 100 checkpoints, 88 reproduced every DS2 prediction exactly. The remaining 12, all CNN+Focal, differed in 1 to 4 of 49,684 beats, which we attribute to floating-point differences of GPU inference. With beat-to-beat intervals, the Macro-F1 of every model rose slightly, in all 20 seeds. The mean increase was $+0.002$ to $+0.005$ across the five models (PC-WavKAN: $0.357 \rightarrow 0.360$), and PC-WavKAN’s V-recall changed by -0.001 . The primary comparison was unchanged: all Holm-adjusted $p \geq 0.38$, and every paired interval still includes zero. The artefact therefore did not inflate the reported performance and does not affect the comparative conclusions. The models were trained with it, and whether training on beat-to-beat intervals would change them was not tested.

Table 8: Parameter-space Prior Retention (4), mean $\pm$ std over 20 models, with mean absolute per-edge deviations from the prior values. Untrained rows use 20 initialisation seeds. The trained isotropic row uses the 20 checkpoints of the no-PCWI ablation arm (base configuration).

Model state	PPR	$ \overline{\Delta\mu} $	$ \overline{\Delta\gamma} $
Untrained, PCWI	0.9729 ± 0.0001	0.013	0.004
Trained, PCWI (PC-WavKAN)	0.7826 ± 0.0351	0.064	0.055
Untrained, isotropic	0.6698 ± 0.0008	0.161	0.051
Trained, isotropic	0.5794 ± 0.0173	0.177	0.080

4.7. Retention of the Prior Values

Table 8 evaluates PPR against its three reference points.

Training moves the PCWI parameters substantially: PPR falls from 0.973 to 0.783, and the mean per-edge translation deviation grows fivefold. The trained PCWI basis nonetheless remains closer to the prior values than an untrained isotropic basis (0.783 against 0.670). The trained isotropic row decides the interpretation. When training starts from the isotropic initialisation, it moves the parameters *further* from the prior values ($0.670 \rightarrow 0.579$). Both comparisons separate the 20 models completely (two-sided Mann-Whitney $p = 7 \times 10^{-8}$). The prior values are therefore not where gradient descent leads. The trained PCWI model’s proximity to them reflects where it started, not a learned preference. Together with the exchangeability of the channel blocks (§3.3), this means the wavelet parameters of this model do not support a physiological interpretation. They support the modest statement that the initial structure is partly retained. Fig. 4 shows the trained functions of one checkpoint against the initial priors. Automated interpretability scores can rate untrained models as highly as trained ones [30], and our results show that parameter-space scores for KANs need the same reference points.

4.8. Cross-Database Evaluation

Table 9 evaluates the five models trained on MIT-BIH on two further databases, forward pass only, 20 seeds each. The first is the St Petersburg INCART 12-lead Arrhythmia Database [25], all 75 records, lead II. The second is the MIT-BIH Supraventricular Arrhythmia Database (SVDB) [31], 78 records, channel 0, whose leads are not labelled. Both were resampled to 360 Hz and then passed through the same extraction code as the training data. A regression test confirms that this code reproduces the training arrays

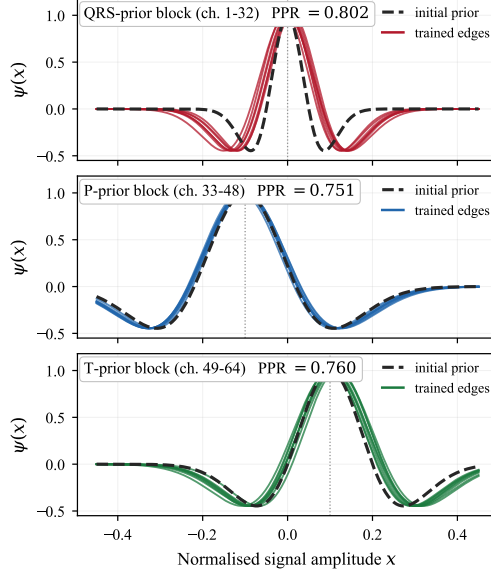


Figure 4: Trained wavelet functions of one checkpoint (seed 42) for each channel block, against the prior each block was initialised at (dashed). Annotated PPR values are for this checkpoint and differ slightly from Table 8’s 20-seed means.

Table 9: Zero-shot cross-database evaluation under the training preprocessing, 20 seeds per model. d_z and p compare PC-WavKAN with each baseline on Macro-F1 (positive favours PC-WavKAN), Holm-adjusted within each database.

Model	INCART (175,785 beats)					SVDB (184,486 beats)				
	Macro-F1	d_z / Holm p	V-Rec	S-Rec		Macro-F1	d_z / Holm p	V-Rec	S-Rec	
PC-WavKAN	0.390 ± 0.017	—	0.832	0.671		0.275 ± 0.012	—	0.788	0.126	
ResNet1D	0.431 ± 0.017	$-1.80 / < 0.001$	0.849	0.760		0.350 ± 0.026	$-2.49 / < 0.001$	0.832	0.256	
Transformer	0.398 ± 0.032	$-0.24 / 0.245$	0.852	0.817		0.316 ± 0.031	$-1.36 / < 0.001$	0.820	0.305	
CNN+Focal	0.431 ± 0.020	$-1.53 / < 0.001$	0.807	0.560		0.354 ± 0.013	$-4.86 / < 0.001$	0.769	0.179	
B-Spline KAN	0.367 ± 0.017	$+0.83 / 0.003$	0.820	0.606		0.267 ± 0.016	$+0.33 / 0.231$	0.791	0.125	

480 byte for byte. Neither database was used for any training, tuning or selection
481 decision. SVDB was assembled to enrich supraventricular ectopy and serves
482 as an S-class stress test. The analysis is exploratory.

483 PC-WavKAN shows no transfer advantage. On INCART its Macro-
484 F1 of 0.390 is significantly below ResNet1D and CNN+Focal (both 0.431;
485 $d_z \leq -1.53$, Holm $p < 0.001$). It is indistinguishable from the Transformer
486 and above B-Spline KAN ($d_z = +0.83$, Holm $p = 0.003$). On SVDB it
487 is significantly below all three non-KAN baselines (Holm $p < 0.001$) and

Table 10: Measured inference, PC-WavKAN, x86 CPU, single thread, PyTorch 2.6.

Metric	Value
Model size, FP32 / INT8 dynamic	0.596 / 0.551 MB (1.08 $\times$)
Latency, batch 1, FP32	2.93 ± 0.38 ms/beat (median 2.78)
Latency, batch 1, INT8 dynamic	4.00 ± 0.35 ms/beat (0.73 $\times$, slower)
Latency, batch 64, FP32	0.429 ms/beat
Peak resident memory	235 MB

indistinguishable from B-Spline KAN, and the two KAN-family models have the lowest Macro-F1 of the five.

On SVDB, both KAN-family models also have the highest S $\rightarrow$ V confusion rates (0.431 and 0.451, mean over seeds, against 0.231–0.395 for the other models) and assign the largest share of beats to V (18.7% and 20.7%, against 8.4–17.6%). We report this as a descriptive pattern; the study does not identify its cause.

Sensitivity to preprocessing. An earlier evaluation of the same checkpoints used windows from only 30 INCART records, on lead I, and applied no filter. It ranked PC-WavKAN first on INCART (Macro-F1 0.374 against 0.316–0.365 for the baselines). Matching the preprocessing to the training data reversed that ranking. Cross-database comparisons of this kind are therefore sensitive to preprocessing details that are rarely reported, and we report only the matched evaluation.

4.9. Efficiency

Table 10 reports inference on an x86 CPU, single-threaded, for the seed-42 checkpoint. Values are the mean of three repeated runs of 1000 timed trials after a 20-iteration warm-up, and $\pm$ is the spread across runs.

Dynamic INT8 quantisation brings no benefit and slows inference. PyTorch’s dynamic quantisation converts `nn.Linear` modules only, while most of this model’s parameters sit in the custom wavelet-KAN layer and the GRU. The FP32 latency supports continuous offline processing on a general-purpose CPU. We made no ARM or on-device measurement and make no claim about wearable deployment.

512 5. Discussion

513 *What the study establishes.* Under a matched protocol with multiplicity
 514 control, the wavelet KAN is not statistically distinguishable from four standard
 515 architectures on Macro-F1, with every paired interval within ± 0.03 . It is not
 516 more accurate than a 30K-parameter Transformer, and we do not present
 517 the wavelet-KAN basis as an accuracy mechanism. Among its design choices,
 518 initialising the wavelet parameters in a structured, prior-centred way helped
 519 relative to an isotropic initialisation, and the choice of mother wavelet did not
 520 measurably matter. The inspectability argument for wavelet KANs does not
 521 transfer to a physiological reading of this model’s parameters. The channel
 522 blocks are exchangeable, and training moves unconstrained parameters away
 523 from the prior values, not towards them. An inspectable parameter is not
 524 thereby an interpretable one. A claim that such parameters encode physiology
 525 would need behavioural evidence, for example a selective loss of one class
 526 when one block’s functions are perturbed. The null models we report are the
 527 minimum needed to interpret a parameter-space score.

528 *Why self-attention over interval features fails.* Removing self-attention from
 529 the rhythm encoder improved validation Macro-F1 with a large effect, while
 530 §4.6 shows the interval features carry real signal. The problem is therefore the
 531 encoder, not the features. Self-attention is close to permutation-equivariant
 532 without positional information, whereas the information in the interval vector
 533 is concentrated in one *position*, RR_0 . An MLP over a fixed-length vector
 534 has that positional distinction for free. We observed this in one architecture
 535 and one dataset. It is a caution worth checking for short fixed-length feature
 536 vectors, not a general result.

537 *The external evaluation qualifies the parity.* Under preprocessing matched
 538 to training, the wavelet KAN transferred worse than both convolutional
 539 baselines on both external databases, and the two KAN-family models were
 540 the weakest on SVDB. In this study, therefore, the parameterisation that is
 541 inspectable in principle came with no accuracy advantage in distribution and
 542 a disadvantage out of distribution. Anyone choosing an architecture for data
 543 from other sources should weigh the second against the first. The earlier,
 544 unmatched evaluation that appeared to show the opposite on INCART is
 545 a concrete example of how a preprocessing mismatch can manufacture an
 546 apparent transfer advantage.

547 *5.1. Limitations*

548 **(1) Parity, not advantage or equivalence.** No equivalence margin was
 549 set. The intervals bound a true Macro-F1 difference at roughly ± 0.03 .

550 **(2) Absolute minority-class performance.** S-recall is 0.198 at precision
 551 0.175. F-recall is negligible in every configuration (0.0003 to 0.0061), reflecting
 552 28 training beats. Q, with 6 training beats, supports no per-class conclusion
 553 and adds a near-constant zero to the five-class average.

554 **(3) Selection history of the test set.** The adopted configuration is
 555 determined by validation data, but DS2 metrics of non-adopted variants were
 556 computed during development.

557 **(4) Scope of the component evidence.** Every component ablation, in-
 558 cluding PCWI, was run on the base configuration. The adopted configuration
 559 differs from it only in the rhythm encoder, and no other component was
 560 re-ablated on it. We do not claim that PCWI improves the adopted model.

561 **(5) Validation partition.** Validation comprises four records, and one record
 562 holds 90% of DS1’s fusion beats. Selection decisions and the validation
 563 reversal in Table 6 should be read with that in mind.

564 **(6) Non-causal inputs and an annotation-derived artefact.** The
 565 inputs include two following intervals, a window that often contains the next
 566 beat, and a zero-phase record-level filter. The interval features also carry an
 567 annotation-derived artefact in 9.2% of DS2 beats. Its effect is bounded in
 568 §4.6. It was not removed by retraining.

569 **(7) External data.** Cross-database evidence rests on two databases. IN-
 570 CART is evaluated on lead II rather than MIT-BIH’s modified lead II, and
 571 SVDB’s leads are unlabelled. PTB-XL [32] was processed but is excluded.
 572 Its labels are record-level, not beat-level. Our extraction used a symmetric
 573 window with the R-peak at sample 180 rather than 90, no filter, R-peaks
 574 from a peak detector rather than expert annotations, and interval features
 575 expressed as ratios to a local 10-beat mean rather than in seconds. A Macro-
 576 F1 of 0.200 ± 0.005 was obtained, and we attribute it to these mismatches
 577 rather than to transfer.

578 **(8) Baselines and hardware.** The B-Spline KAN baseline is a single-bump
 579 variant, not a grid-based spline KAN. Latency is x86 CPU only.

580 *Future work.* The results point to three specific questions: whether a causal
 581 variant (preceding intervals only, a truncated window and a causal filter)
 582 preserves the parity; whether the wavelet-KAN family’s behaviour holds on

583 12-lead recordings with beat-level labels; and whether any behavioural test
584 links individual wavelet parameters to class-specific evidence.

585 6. Conclusion

586 We evaluated a 153,045-parameter wavelet Kolmogorov–Arnold network for
587 inter-patient ECG heartbeat classification under the de Chazal protocol, with
588 20 paired seeds, multiplicity correction, effect sizes, confidence intervals and
589 architecture selection on validation data. Its DS2 Macro-F1 of 0.357 ± 0.019
590 is not statistically distinguishable from four standard baselines of 29K–118K
591 parameters, and all five models share a poor supraventricular operating
592 point. Structured initialisation of the wavelet parameters helped relative
593 to an isotropic one, the mother wavelet did not measurably matter, and
594 self-attention over interval features was harmful. The parameters do not
595 support a physiological reading once exchangeability and a trained null are
596 considered. Class-balanced sampling raised DS2 supraventricular recall but
597 not on the validation partition. Under preprocessing matched to training, the
598 wavelet KAN transferred worse than both convolutional baselines to INCART
599 and SVDB. We report these null and negative findings alongside the positive
600 ones because, on this evidence, they are the more transferable part of what
601 the study established.

602 CRedit authorship contribution statement

603 **Venkateramanan Manivannan:** Conceptualization, Methodology, Soft-
604 ware, Validation, Formal analysis, Investigation, Data curation, Writing –
605 original draft, Visualization. **Ramanathan Lakshmanan:** Conceptualiza-
606 tion, Resources, Validation, Supervision, Project administration, Funding
607 acquisition, Writing – review & editing.

608 Declaration of competing interest

609 The authors declare that they have no known competing financial interests
610 or personal relationships that could have appeared to influence the work
611 reported in this paper.

612 Funding

613 The Vellore Institute of Technology (VIT) funds the article processing
614 charge. This research received no other specific grant.

615 **Ethics statement**

616 This study uses only publicly available, de-identified PhysioNet databases
617 [25]. No new human-subject data were collected, and no institutional review
618 board approval was required.

619 **Data availability**

620 All datasets are publicly available from PhysioNet [25]: the MIT-BIH
621 Arrhythmia Database [24] (<https://physionet.org/content/mitdb/1.0.0/>), the St Petersburg INCART 12-lead Arrhythmia Database (<https://physionet.org/content/incartdb/1.0.0/>), the MIT-BIH Supraventricular Arrhythmia Database [31] (<https://physionet.org/content/svdb/1.0.0/>) and PTB-XL [32] (<https://physionet.org/content/ptb-xl/1.0.3/>).

626 **Code availability**

627 <https://github.com/vrhrsr/WavKAN-CL> contains the source code, pre-
628 processing and training scripts, statistical and figure code, the per-seed
629 records of every arm reported here (test metrics, training histories, confusion
630 matrices), and the 20 checkpoints of the evaluated PC-WavKAN config-
631 uration. Baseline checkpoints are available from the authors on request.
632 `src/verify_manuscript_numbers.py` re-derives the quantitative claims of
633 this manuscript from those files. The repository name is historical.

634 **Declaration of generative AI and AI-assisted technologies in the**
635 **manuscript preparation process**

636 During the preparation of this work the authors used Claude (Anthropic)
637 in order to assist with drafting and revising the manuscript text, writing
638 and auditing analysis, verification and figure-generation code, and checking
639 reported numbers against the result files. After using this tool, the authors
640 reviewed and edited the content as needed and take full responsibility for the
641 content of the published article.

642 References

- 643 [1] J. Takalo-Mattila, J. Kiljander, J.-P. Soininen, Inter-patient ECG clas-
644 sification using deep convolutional neural networks, Proceedings of the
645 21st Euromicro Conference on Digital System Design (DSD) (2018)
646 421–425doi:10.1109/DSD.2018.00077.
- 647 [2] L. Guo, G. Sim, B. Matuszewski, Inter-patient ECG classification with
648 convolutional and recurrent neural networks, Biocybernetics and Biomed-
649 ical Engineering 39 (3) (2019) 868–879. doi:10.1016/j.bbe.2019.06.
650 001.
- 651 [3] T. Akan, S. Alp, M. A. N. Bhuiyan, ECGformer: A transformer-based
652 ensemble technique for cardiac arrhythmia classification, arXiv preprint
653 arXiv:2401.05434 (2024).
- 654 [4] R. Bahrami, A. M. Fotouhi, Investigation of inter-patient, intra-patient,
655 and patient-specific based training in deep learning for classification of
656 heartbeat arrhythmia, Cardiovascular Engineering and Technology 16
657 (2025) 333–342. doi:10.1007/s13239-025-00777-y.
- 658 [5] Q. Xiao, K. Lee, S. A. Mokhtar, I. Ismail, A. L. b. M. Pauzi,
659 Q. Zhang, P. Y. Lim, Deep learning-based ECG arrhythmia classi-
660 fication: A systematic review, Applied Sciences 13 (8) (2023) 4964.
661 doi:10.3390/app13084964.
- 662 [6] G. A. L. Silva, et al., A systematic review of ECG arrhythmia classifica-
663 tion: Adherence to standards, fair evaluation, and embedded feasibility,
664 arXiv preprint arXiv:2503.07276 (2025).
- 665 [7] P. De Chazal, M. O’Dwyer, R. B. Reilly, Automatic classification of
666 heartbeats using ECG morphology and heartbeat interval features, IEEE
667 Transactions on Biomedical Engineering 51 (7) (2004) 1196–1206. doi:
668 10.1109/TBME.2004.827359.
- 669 [8] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y.
670 Hou, M. Tegmark, KAN: Kolmogorov–Arnold networks, arXiv preprint
671 arXiv:2404.19756 (2024).
- 672 [9] Z. Bozorgasl, H. Chen, Wav-KAN: Wavelet Kolmogorov-Arnold networks,
673 arXiv preprint arXiv:2405.12832 (2024).

- [10] C. Zhao, B. Lai, Y. Xu, Y. Wang, H. Dong, MAK-Net: A multi-scale attentive Kolmogorov–Arnold network with BiGRU for imbalanced ECG arrhythmia classification, *Sensors* 25 (13) (2025) 3928. doi:10.3390/s25133928.
- [11] N. Kim, W. Seo, J.-h. Kim, S. Y. Choi, S.-M. Park, WavelNet: A novel convolutional neural network architecture for arrhythmia classification from electrocardiograms, *Computer Methods and Programs in Biomedicine* 231 (2023) 107375. doi:10.1016/j.cmpb.2023.107375.
- [12] F.-y. Zhou, Y.-h. Sun, Y.-w. Wang, Inter-patient ECG arrhythmia heartbeat classification network based on multiscale convolution and FCBA, *Biomedical Signal Processing and Control* 89 (2024) 105789. doi:10.1016/j.bspc.2023.105789.
- [13] S. Mousavi, F. Afghah, Inter- and intra-patient ECG heartbeat classification for arrhythmia detection: A sequence to sequence deep learning approach, in: 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2019, pp. 1308–1312. doi:10.1109/ICASSP.2019.8683140.
- [14] T. Li, Z. Zhao, C. Sun, L. Cheng, X. Chen, R. Yan, R. X. Gao, WaveletKernelNet: An interpretable deep neural network for industrial intelligent diagnosis, *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 52 (4) (2022) 2302–2312. doi:10.1109/TSMC.2020.3048950.
- [15] S. Rigas, D. Verma, G. Alexandridis, Y. Wang, Initialization schemes for Kolmogorov-Arnold networks: An empirical study, in: International Conference on Learning Representations (ICLR), 2026. arXiv:2509.03417.
- [16] M. Buda, A. Maki, M. A. Mazurowski, A systematic study of the class imbalance problem in convolutional neural networks, *Neural Networks* 106 (2018) 249–259. doi:10.1016/j.neunet.2018.07.011.
- [17] B. Kang, S. Xie, M. Rohrbach, Z. Yan, A. Gordo, J. Feng, Y. Kalantidis, Decoupling representation and classifier for long-tailed recognition, in: International Conference on Learning Representations (ICLR), 2020. arXiv:1910.09217.

- 706 [18] Y. Bengio, J. Louradour, R. Collobert, J. Weston, Curriculum learning, in:
707 Proceedings of the 26th International Conference on Machine Learning
708 (ICML), ACM, Montreal, Canada, 2009, pp. 41–48. doi:10.1145/
709 1553374.1553380.
- 710 [19] J. Bae, J. Shin, Handling imbalanced medical dataset with continuous
711 class features using improved contrastive learning, in: Proceedings of the
712 IEEE International Conference on Consumer Electronics (ICCE), IEEE,
713 2025.
- 714 [20] S. Schmale, P. Hempel, N. Spicher, Curriculum learning using real and
715 simulated data in deep learning models for electrocardiography classifi-
716 cation, in: Proceedings of the 47th Annual International Conference of
717 the IEEE Engineering in Medicine and Biology Society (EMBC), IEEE,
718 2025. doi:10.1109/EMBC58623.2025.11252863.
- 719 [21] J. Dai, Q. Zhang, C. Dai, D. Lan, C. Han, Angular gaussian super-
720 vised contrastive learning for long-tailed electrocardiogram arrhythmia
721 diagnosis, arXiv preprint arXiv:2607.14613 (2026).
- 722 [22] M. M. Farag, A tiny matched-filter-based CNN for inter-patient ECG
723 classification and arrhythmia detection at the edge, Sensors 23 (3) (2023)
724 1365. doi:10.3390/s23031365.
- 725 [23] D. H. Elsheikhy, A. S. Hassan, N. M. Yhiea, A. M. Fareed, E. A. Rashed,
726 Lightweight deep learning architecture for multi-lead ECG arrhythmia
727 detection, Sensors 25 (17) (2025) 5542. doi:10.3390/s25175542.
- 728 [24] G. B. Moody, R. G. Mark, The impact of the MIT-BIH Arrhythmia
729 Database, IEEE Engineering in Medicine and Biology Magazine 20 (3)
730 (2001) 45–50. doi:10.1109/51.932724.
- 731 [25] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C.
732 Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, H. E.
733 Stanley, PhysioBank, PhysioToolkit, and PhysioNet: Components of
734 a new research resource for complex physiologic signals, Circulation
735 101 (23) (2000) e215–e220. doi:10.1161/01.CIR.101.23.e215.
- 736 [26] E. J. d. S. Luz, W. R. Schwartz, G. Cámara-Chávez, D. Menotti,
737 ECG-based heartbeat classification for arrhythmia detection: A survey,

- 738 Computer Methods and Programs in Biomedicine 127 (2016) 144–164.
739 doi:10.1016/j.cmpb.2015.12.008.
- 740 [27] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics
741 Bulletin 1 (6) (1945) 80–83. doi:10.2307/3001968.
- 742 [28] S. Holm, A simple sequentially rejective multiple test procedure, Scandi-
743 navian Journal of Statistics 6 (2) (1979) 65–70.
- 744 [29] J. Cohen, Statistical Power Analysis for the Behavioral Sciences, 2nd
745 Edition, Lawrence Erlbaum Associates, Hillsdale, NJ, 1988.
- 746 [30] T. Heap, T. Lawson, L. Farnik, L. Aitchison, Automated interpretability
747 metrics do not distinguish trained and random transformers, arXiv
748 preprint arXiv:2501.17727 (2025).
- 749 [31] S. D. Greenwald, Improved detection and classification of arrhythmias in
750 noise-corrupted electrocardiograms using contextual information, Ph.D.
751 thesis, Harvard–MIT Division of Health Sciences and Technology (1990).
- 752 [32] P. Wagner, N. Strodthoff, R.-D. Bousseljot, D. Kreiseler, F. I. Lunze,
753 W. Samek, T. Schaeffter, PTB-XL, a large publicly available electro-
754 cardiography dataset, Scientific Data 7 (1) (2020) 154. doi:10.1038/
755 s41597-020-0495-6.